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LLM-enhanced causal graph learning for real-time crash risk prediction

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ABSTRACT

Accurate and reliable risk prediction is crucial for preventing traffic crashes and improving road safety. However, existing approaches often struggle to balance predictive performance with causal interpretability. To bridge this gap, we propose a closed-loop framework that integrates causal discovery, semantic enhancement, and spatio-temporal prediction. First, transfer entropy extracts event-specific causal relationships from dangerous driving scenarios to construct initial causal graphs. These graphs are then refined using a GPT-2-based language model fine-tuned with Low-Rank Adaptation (LoRA), which employs a graph enhancement strategy to reinforce salient causal pathways. Finally, the semantically enhanced graphs are fused with temporal driving features using a spatio-temporal model that combines Graph Attention Networks (GAT) and Long Short-Term Memory (LSTM) networks for real-time risk prediction. Evaluated on car-following scenarios from the highD dataset, our framework achieves an accuracy of 0.956, F1-score of 0.872, and AUC of 0.985, significantly outperforming traditional baselines. Ablation studies further confirm that the GPT-2-based causal enhancement with LoRA fine-tuning is a significant contributor to the model's predictive accuracy. These findings indicate that our framework offers a powerful solution for enhancing both precision and interpretability in collision risk prediction, holding strong potential for deployment in real-time traffic management systems.

1. Introduction

Road traffic accidents remain a critical global safety concern. In 2023, they caused approximately 1.19 million deaths and led to economic losses equivalent to about 3% of global GDP (Organization, 2023).Many studies have shown that the post-crash evaluation and infrastructure reconstruction are effective in reducing the frequency of traffic s crashes (Greaves & Ellison, 2011; S & Ray, 2012). Although these passive safety interventions have saved many lives, they are also insufficient to prevent crashes and reveal the underlying causes of traffic crashes.

In recent years, proactive safety strategies have attracted increasing attention, with driving risk prediction identified as a key research direction. By forecasting potential collision risks and implementing targeted interventions, these strategies have been shown to significantly reduce the likelihood of collisions. For example, Q. Li et al. (2024b) introduced an the International Road Assessment Program (IRAP) model to support infrastructure upgrades, potentially preventing 160,000 fatalities annually. Cicchino (2017) reported that forward collision warning with automatic emergency braking reduced rear-end crashes by 50%. Similarly, Lubbe (2017) developed a real-time system capable of issuing warnings up to 0.8 s before a crash. These findings highlight the importance of developing driving risk prediction methods to improve road safety.

As risk prediction is fundamental to proactive safety and has become a central topic in road safety research, numerous methods have been developed to improve prediction accuracy. Previous studies employed machine learning models, such as Artificial Neural Networks (ANN), Support Vector Machines (SVM) and logistic regression, to estimate crash risk (Celik & Oktay, 2014; Yu & Abdel-Aty, 2013). More recently, deep learning models, including Convolutional Neural Networks (CNN), Deep Neural Networks (DNN), Long Short-Term Memory (LSTM), and Graph Attention Networks (GATs), have achieved superior performance (Sharma et al., 2023; Shi et al., 2022). Based on this, recent efforts have explored hybrid frameworks that integrate deep learning with additional components. For example, Shi et al. (2022) combined CNNs and Recurrent Neural Networks (RNNs) to capture high-frequency temporal features, followed by eXtreme Gradient Boosting (XGBoost) for secondary risk classification. Gao et al. (2025) proposed STCAGNN-RNKDE framework, which integrates GNN with Ripley's K function and kernel

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density estimation to explore spatial diffusion of multi-source risks. Despite these improvements, current deep learning-based models still lack explicit representations of the causal relationships between driving behavior, traffic context and risk outcomes, limiting their interpretability and feasibility for safety interventions.

To enhance the interpretability of prediction models, recent studies have integrated causal inference into risk prediction frameworks. For example, Zaidi et al. (2025) used a causal forest to evaluate freeway exit improvements and found substantial variation in treatment effects across roadway conditions, highlighting the need for context-specific causal modeling. Zhao (2021) suggested that Bayesian network–based causal framework can effectively capture the event-level heterogeneity. While event-level causal analysis offers a promising way to reveal these mechanisms, it still faces major challenges in practice. First, causal structures built on static rules often fail to distinguish critical causal pathways, making it difficult to guide graph optimization or support predictive modeling (Shen et al., 2020).Second, with the increasing incorporation of heterogeneous information, such as multidimensional time series, graph structures and semantic context, traditional methods struggle to construct context-adaptive causal graphs (Zhang et al., 2023). More importantly, it remains challenging to balance causal structures and prediction tasks, which hinders the ability to enhance both model performance and interpretability simultaneously.

In recent years, the emergence of Large Language Models (LLMs) has provided a promising tool to solve these challenges. Unlike traditional methods constrained by predefined causal structures, LLMs possess powerful contextual reasoning and pattern recognition capabilities, which can bridge the gap between data-driven prediction and causal understanding (Constantinou et al., 2025; Jiao et al., 2024; Kiciman et al., 2023). Therefore, this study proposes an LLM-based Graph with Spatio-Temporal component (LLMG + ST) framework, which dynamically refines causal graphs and integrates them into a risk prediction model. The key contributions are as follows:

1) Conducting event-level causal analysis by constructing individual causal matric for each dangerous event, rather than a single global structure. It captures the heterogeneity across events, where each event is driven by distinct causal patterns.
2) Integrating LLM to enhance causal graphs in traffic safety. The LLM uses domain knowledge and context to refine causal structures based on discovered relationships.
3) Proposing a closed-loop framework that integrates causal discovery, semantic enhancement, and spatial–temporal prediction within a unified architecture. This framework eliminates the traditional separation between causal inference and predictive tasks.

2. Literature review

2.1. Real-time risk prediction

Traditional risk prediction methods primarily relied on statistical tools, such as extreme value theory (EVT) and conflict indicators, to estimate crash risk. For example, Wang et al. (2018) employed EVT to surrogate safety measures and estimate crash probabilities. Similarly, Zheng et al. (2018) applied a two-variable EVT model to describe both Post Encroachment Time (PET) and Time-To-Collision (TTC), providing a clearer picture of potential risks. While these methods provided a reliable basis for traffic conflict analysis, they are limited in capturing complex interactions and dynamic driver behaviors in traffic environments (Ali et al., 2023).

Furthermore, researchers have begun to explore deep learning–based models. Architectures such as LSTM, GRU, and Transformer are used to capture temporal patterns in driving behavior, improving prediction accuracy and response time. For example, Hussain et al. (2021) proposed an optimized GRU model for traffic flow prediction, improving accuracy and response speed. Yuan et al. (2019) applied LSTM for real-time crash risk prediction, achieving higher sensitivity by memorizing long histories. Guo et al. (2023) developed a Transformer-based network that captures temporal dependencies in driving behavior. Despite these progress, previous models often treat driving behavior as purely time-dependent sequences. They neglect spatial interactions and struggle to capture the complex dynamics between multiple vehicles.

To address these limitations, graph based spatial models have received increasing attention. In such frameworks, vehicles are encoded as nodes, while their relative distances and speed differences are encoded as edges, allowing the model to capture spatial–temporal dynamics. For example, Gao et al. (2024) proposed a probabilistic graph neural network with uncertainty, which improves prediction accuracy and provides confidence intervals. Yu et al. (2021) developed a deep spatiotemporal graph convolutional network (DSTGCN) that combines CNNs and graph convolutions for city-scale risk modeling. Liu et al. (2023) introduced attention mechanisms and focal loss to integrate multi-source data, including traffic flow, weather and road features, improving crash risk prediction for urban road networks. Han et al. (2024) introduced a time-decay Transformer to model the spatio-temporal effects of abnormal driving behavior, achieving higher accuracy with low complexity. Gao et al. (2024) further developed an uncertainty-aware spatiotemporal graph neural network to predict multi-step crash risks at the road level. Although these models improve accuracy and adaptability, their black-box nature and reliance on correlations present major challenges. By capturing statistical patterns without understanding underlying causal mechanisms, they often exhibit poor generalization and limited interpretability on out-of-sample data (Mannering et al., 2020).

Recently, some studies have attempted to integrate causal inference with deep learning methods. S. Li et al. (2024c) first identified confounders using a Conditional Shapley Value Index (CSVI), and then applied Doubly Robust Learning to infer crash impacts. Chen et al. (2022) employed optimized structure learning to construct a Bayesian Network for causal discovery, then conducting risk prediction based on the causal relationships. Lyu et al. (2025) applied Structural Agnostic Model (SAM) algorithm to uncover causal relationships among conflict-related variables, and then used GATs to predict real-time conflict risks using causally structured data. However, these studies primarily adopt a two-stage causal framework, where causal discovery and risk prediction are conducted as separate processes. This separation prevents dynamic interaction between causal structures and predictive feedback, thereby limiting adaptivity and interpretability.

In summary, although deep learning methods have improved the accuracy of traffic risk prediction, they often ignore the underlying causal relation. Moreover, in most current two-stage causal modeling researches, causal graph building and model training are conducted separately, with limited feedback from the prediction task to update the causal structure. Therefore, it is essential to jointly improve both prediction performance and causal interpretability.

2.2. Causal discovery methods

In recent years, causal discovery methods have received increasing attention in the field of traffic safety. Previous studies primarily relied on statistical approaches, with Granger causality tests being among the most widely used. This method infers causal relationships by evaluating whether one time series can predict another. For example, Ouni and Mraihi (2025) combined the Granger causality test with a dynamic autoregressive distributed lag (DARDL) model to reveal the long-term causal effects on traffic fatality rates in Tunisia. Huang et al. (2025) employed Granger-based analysis to construct a hierarchical causal path for traffic accidents, identifying weather and speed changes as key driving factors contributing to fatal crashes. However, because the Granger is built on the assumption of linear relationships, it struggles to capture the nonlinear dependencies commonly present in complex traffic systems, which limits both its performance and robustness.

To overcome the limitations of linear assumptions, information-theoretic methods have been introduced into causal inference. Among them, Transfer Entropy (TE) has been widely applied to capture nonlinear and directional information flow between variables(Yang et al., 2021). For example, J. Li et al. (2024a) applied multivariate TE to capture causality in delay propagation under dynamic conditions. Reddy et al. (2025) integrated entropy-based mechanisms into image modeling, enhancing the detection of nonlinear causal paths in distracted driving. Although TE method provides theoretical advantages in capturing complex dependencies, its practical application is limited by noise sensitivity and high computational cost.

Unlike TE methods that that focus on pairwise causal relationships, structure learning methods intend to infer entire causal graph from data. For example, Jin et al. (2025) employed SAM method to discover causal relations in tunnel collisions. Wang et al. (2024) employed the additive noise model (ANM) to analyze the causal mechanisms between mixed traffic flow and highway surface deterioration. Mao et al. (2024) employed convergent cross-mapping (CCM) method to detect the causal propagation directions in multi-point speed data. Although these methods demonstrate clear advantages in accuracy and precision, they still face challenges, including poor interpretability and insufficient prior knowledge.

To balance prediction accuracy and interpretability, incorporating domain knowledge into risk causal discovery has become a promising direction. By integrating physical constraints and empirical knowledge from text, this approach enables a deeper integration between structured data and domain knowledge. For example, Wang et al. (2023) combined grounded theory with Bayesian networks to model hazardous chemical transport risks. Similarly, Zou and Yue (2017) introduced expert knowledge into Bayesian networks, achieving better stability and causal consistency. Compared with traditional approaches, expert-informed causal graphs enhance both prediction performance and model interpretability.

The recent LLMs have emerged as powerful tools for constructing dynamic domain-informed causal graphs, with potential benefits for risk prediction. For example, Constantinou et al. (2025) employed GPT-4 to generate candidate causal pathways, which were then adjusted using graph optimization algorithms. Dai et al. (2025) proposed a causal inference framework that integrates generalized causal forests (GRF) with LLMs to evaluate the safety impacts of dedicated bicycle lanes. Additionally, Zhou et al. (2024) developed CausalKGPT, a causal knowledge-enhanced LLM for analyzing quality issues in aerospace manufacturing. These studies proved that LLMs can dynamically adapt to the contextual differences of individual traffic events, identify semantically important causal chains and establish bidirectional mappings between structured data and causal knowledge.

To summarize, although studies on causal discovery have achieved significant progress, the integration of domain knowledge with causal inference approaches remain limited. Moreover, the application of causal discovery in high-risk traffic scenarios is still in its infancy. To address these challenges, this study proposes a risk prediction framework that combines prior knowledge with adaptive learning strategies. Furthermore, LLMs are employed to refine these causal structures through semantic reasoning, improving both interpretability and accuracy under high-risk traffic conditions.

3. Methodology

3.1. Research framework

As shown in Fig. 1, our proposed framework consists of three main components. The first component is the generation of an initial causal graph. We first identify dangerous events from the HighD dataset by selecting scenarios with a TTC below 3 s. Then, the Transfer Entropy algorithm is adopted to construct an initial causal graph for each event. The second component is the enhancement of the causal graph. A Causal Enhancement Agent, which is based on a GPT-2 backbone and fine-tuned with LoRA, is employed to refine the initial graph structure. This agent integrates semantic information, including safety rules, driving knowledge and highway context, to construct an enhanced causal graph. The third component conducts real-time collision risk prediction. This component uses a hybrid model that combines GAT model and LSTM model through a feature fusion layer for predicting the real-time risk. In addition, an update loop is incorporated into the framework to improve the causal enhancement agent.

3.2. Transfer entropy method

To quantify complex nonlinear dependencies in time series, Schreiber (2000) initially proposed TE concept. Since TE does not rely on assumptions about underlying data-generating mechanisms, it is suited for capturing dynamic and nonlinear information transfer between signals. The formula is defined by equation (1).

$$T_{x\to y} = \sum\nolimits_{y_{t+1}, y_t^k, x_t^l} p\left(y_{t+1}, y_t^k, x_t^l\right) log_2\left[\frac{p\left(y_{t+1}|y_t^k, x_t^l\right)}{p\left(y_{t+1}|y_t^k\right)}\right] \tag{1}$$

Where, $T_{x\to y}$ denotes the information transmission from x to y, y_{t+1} is the value of y at time $t+1$; $y_t^k = \left(y_t, y_{t-1}, \cdots, y_{t-k+1}\right)$ is the past k-1 values of variable y, $x_t^l = \left(x_t, x_{t-1}, \cdots, x_{t-l+1}\right)$ represents the past $l-1$ values of variable x; $p\left(y_{t+1}, y_t^k, x_t^l\right)$ is the joint probability, while $p\left(y_{t+1}|y_t^k, x_t^l\right)$ and

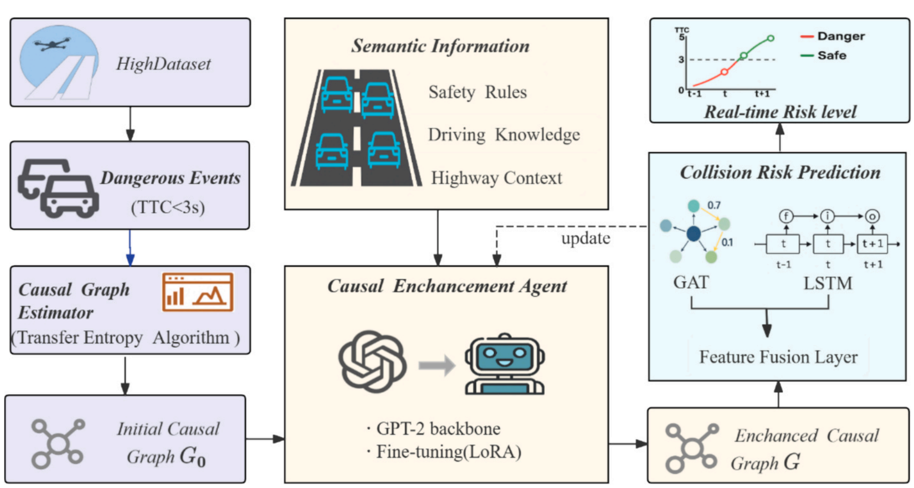


Fig. 1. Research framework of this study.

$p(y_{t+1}|y_t^k)$ are conditional probabilities.

However, time series often contain complex temporal dependencies and the appropriate time-delay length is typically unknown, the original TE may struggle to accurately capture time-lagged causal relationships. To address this issue, variable-delay formulations are introduced to replace fixed-delay assumptions in the original TE. The formula is modified as equation (2).

$$T_{x\to y}(\tau) = \sum\nolimits_{y_{t+1},y_t^k,x_{t-\tau+1}^l} p\left(y_{t+1},y_t^k,x_{t-\tau+1}^l\right) \times lb\left[\frac{p\left(y_{t+1}|y_t^k,x_{t-\tau+1}^l\right)}{p\left(y_{t+1}|y_t^k\right)}\right] \tag{2}$$

Where, τ represents the lag step size. The $lb[\cdot]$ denotes the logarithm with base 2.

Subsequently, the normalized TE sequence is obtained through the systematic substitution of B sequences. For each substitution, the corresponding TE value is recalculated. The computation is defined by equation (3).

$$p_{x\to y} = \frac{1+\sum_{b=1}^{B}\left[T_{x\to y}^{b}(\tau) \geq T_{x\to y}(\tau)\right]}{1+B} \tag{3}$$

Where, $p_{x\to y}$ denotes the permutation-based p-value from x to y.B represents the number of replacement sequences.

Finally, considering the influence of both weak causal links and significance, the maximum normalized value of $T_{x\to y}$ is selected for modification as equation (4).

$$\overline{T}_{x\to y}(\tau) = \begin{cases} T_{x\to y}(\tau) if T_{x\to y}(\tau) > \theta and p_{x\to y} < \alpha \\ 0 else \end{cases} \tag{4}$$

Where, $\overline{T}_{x\to y}(\tau)$ represents the maximum *TE* value from variable x to variable y, θ denotes the minimum threshold for TE values, set at 0.01; and α represents the minimum threshold for $p_{x\to y}$, set at 0.05.

3.3. LLM optimization with LoRA

To satisfy real-time constraints, the proposed model is built upon GPT-2(0.78B parameters) and constructs a unified representation by integrating textual sequences with causal graph embeddings. To enable efficient fine-tuning of the large model, LoRA is incorporated into the framework. Furthermore, the hidden representations are refined through reinforced structural constraint. The overall framework is illustrated in Fig. 2.

The LLM input consists of a structured prompt that encodes domain knowledge across three dimensions. These include safety rules derived from established traffic safety standards, driving knowledge based on physical rules, and highway context extracted from the highD dataset. The complete prompt template is provided in Appendix A. This prompt is tokenized using the GPT-2 tokenizer to produce a sequence of tokens.

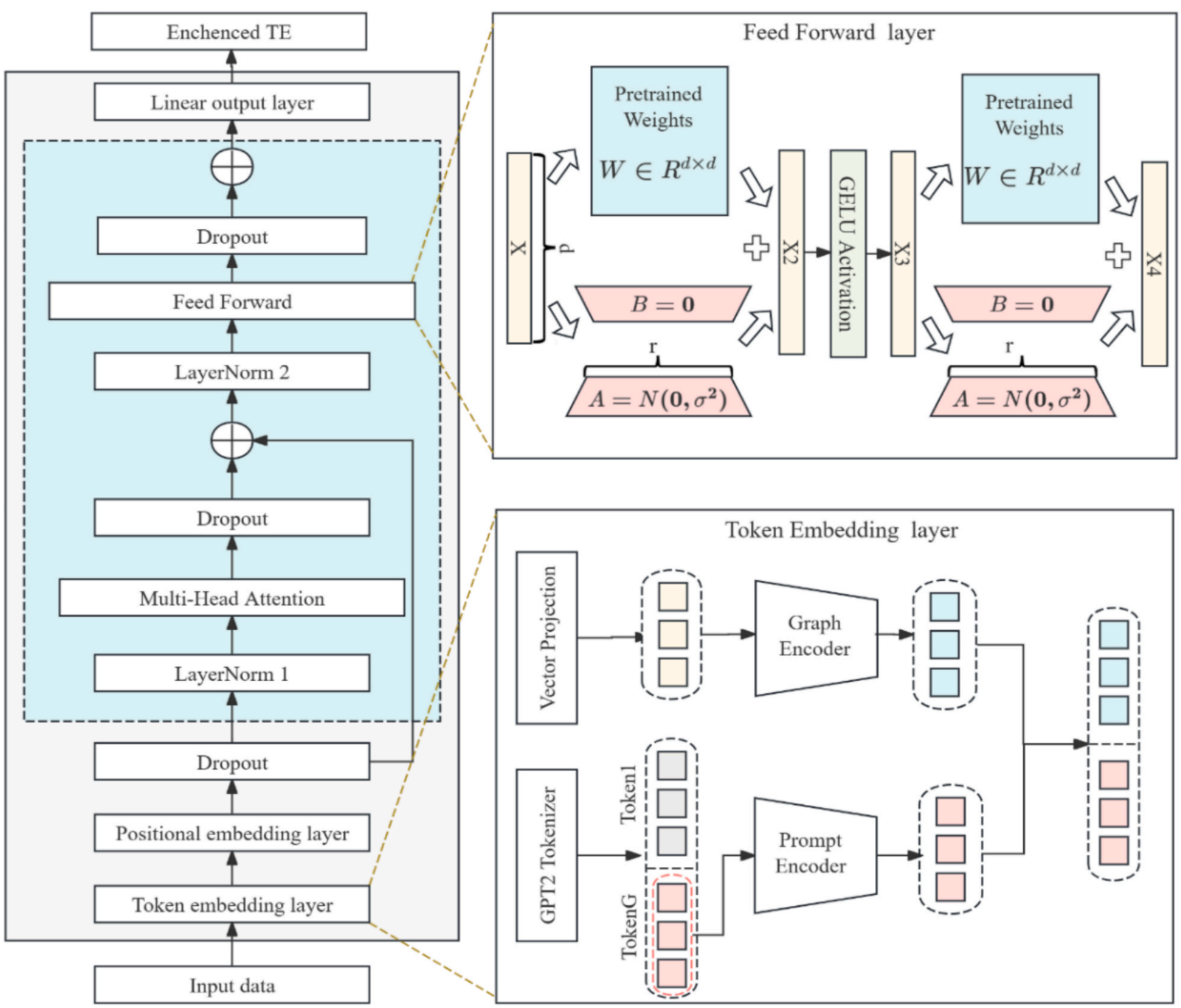


Fig. 2. Diagram of the structure for GPT-2 with LoRA fine-tuning.

These tokens are then projected through a prompt encoder layer, resulting in a sequence of embeddings denoted as $g_{1:t}$ Similarly, the graph structural data is projected into a latent weight space, producing a vector sequence denoted as $e_{1:m}$. These two types of encodings are subsequently integrated to form a unified input matrix, as defined in equation (5).

$$X = [g_{1:t}; e_{1:m}] \tag{5}$$

Where, t represents the sequence length and m denotes the number of nodes.

Then, the input tensor is combined with learnable positional embeddings to produce position-aware representations. These representations are further fed into a stack of l transformer encoder layers, each consisting of a Multi-Head Attention (MHA) sublayer and a Feed-Forward Network (FFN) sublayer. Residual connections are applied around each sublayer to facilitate gradient flow and enhance representation learning. For the l_{th} layer, the attention is computed as defined in equation (6).

$$U^l = H^{l-1} + Dropout\left(MHA\left(LayerNorm1\left(H^{l-1}\right)\right)\right) \tag{6}$$

Where, U^l denotes the input hidden representation at the l layer. H^{l-1} is the input hidden representation at the $l-1$ layer, $MHA(\cdot)$ represents the multi-head self-attention method.$LayerNorm1(\cdot)$ is the first layer normalization applied before the attention sublayer. Subsequently, LoRA is applied at the feedforward layer to enable efficient fine-tuning. The original weight matrix $W \in \mathrm{R}^{d\times r}$ is kept frozen, while trainable low-rank matrices are initialized with $A \sim N(0, \sigma^2)$ and $B = 0$. By implementing incremental updates, both the forward and backward network representations are adjusted as equation (7).

$$FFN_{LoRA}(v) = (W_2 + B_2A_2)\cdot GELU((W_1 + B_1A_1)v) \tag{7}$$

Where, v denotes the input hidden state vector. W_1 and W_2 are frozen weight matrices, while A_1 and A_2 are trainable low-rank adaptation matrices. B_1 and B_2 represent the corresponding projection matrices. $GELU(\cdot)$ is the Gaussian Error Linear Unit activation function. Then the updated hidden state representation is defined in equation (8).

$$H^l = U^l + Dropout\left(FFN_{LoRA}\left(LayerNorm2\left(U^l\right)\right)\right) \tag{8}$$

Where, $LayerNorm2(\cdot)$ denotes the second layer normalization operation applied within the Transformer block. After passing through the transformer stack, the model exports enhanced TE metrics through linear output layers for risk prediction tasks.

3.4. Traffic risk prediction

To capture both the spatio-temporal patterns and structural dependencies in vehicle interactions, a spatial–temporal model is proposed as illustrated in Fig. 3.This model combines LSTM with GAT, where LSTM captures temporal dynamics and GAT encodes graph-structured spatial dependencies. First, trajectory sequences are extracted using a sliding window approach, which is defined by three parameters: window length L, prediction length F, and step size S. The resulting trajectories are denoted as $\mathbf{X}$. These sequences are then processed by the LSTM to capture temporal features. Then, a temporal attention mechanism is applied to assign different weights to each time step. This produces the final temporal representation z_{time}, as defined in equation (9).

$$z_{time} = \sum\nolimits_{t=1}^{T} \alpha_t h_t \tag{9}$$

Where, h_t is the hidden state output from the LSTM at time t, and α_t denotes the corresponding attention weight at time t.

The GPT-2-LoRA enhanced TE matrix is incorporated into the spatial module. Specifically, each trajectory segment uses the causal graph corresponding to its driving event as the structural input. To model both node attributes and edge connections, the GAT method is employed. For

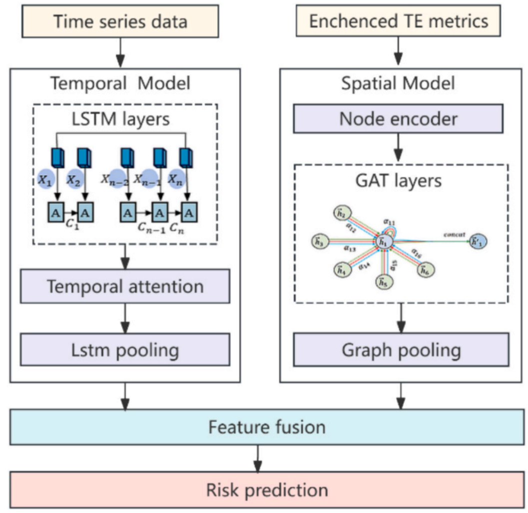


Fig. 3. The framework of LSTM-GAT.

each node pair (i,j) within the standard multi-head GAT framework, the initial edge attention weight is computed. To further improve the robustness of the constructed graph, a splitting function is introduced for the h-th attention head, as defined in equation (10).

$$e_{ij}^{(h)} = \left(a_{src}^{(h)} \bullet W^{(h)}x_i\right) + \left(a_{dst}^{(h)} \bullet W^{(h)}x_j\right) + \left(\beta^{(h)} g\left(w_{ij}^{*}\right)\right) \tag{10}$$

Where,$\mathrm{W}^{(h)}$ denotes the learnable projection matrix for the h attention head. $\mathrm{a}_{\mathrm{src}}^{(h)}$ and $\mathrm{a}_{\mathrm{dst}}^{(h)}$ are the attention vectors for source and target nodes. w_{ij}^{*} denotes the enhanced edge weight produced by GPT-2-LoRA, which only modifies the weights of existing edges without introducing new ones. $\beta^{(h)}$ is a learnable coefficient; and $\mathrm{g}(\cdot)$ denotes a stable transformation function applied to edge weights. To get the final graph representation, the outputs from the stacked GAT layers are aggregated using an attention pooling layer, as defined in equation (11).

$$z_{graph} = READOUT(\mathscr{F}_{GAT}(N, A, W^{*})) \tag{11}$$

Where, $\mathscr{F}_{GAT}(\cdot)$ denotes the stacked multi-head GAT encoder with edge-weight fusion, and $READOUT(\cdot)$ is a graph pooling operator producing the final embedding. N denotes the node feature matrix, A is the adjacency matrix encoding the graph topology, and W^{*} represents the enhanced edge-weight matrix.

To integrate temporal dynamics and causal dependencies, the sequential embedding and the graph embedding are combined through a feature fusion layer. The fused vector is defined as equation (12).

$$z = FUSE\left(z_{time}; z_{graph}\right) \tag{12}$$

Where, $FUSE(\cdot)$ denotes a fusion function. The fused feature is then used for prediction, as defined in equation (13).

$$\hat{y} = f(z) \tag{13}$$

Where, $\mathrm{f}(\cdot)$ is the prediction head. The training objective includes the task loss, key-path enhancement and regularization terms to promote robustness and structural interpretability. The formulation of the key-path enhancement is presented in equation (14), and the overall loss function is defined in equation (15).

$$L_{path} = \sum\nolimits_{i=1}^{n}\left[(\Delta TE_i - \eta)^2 + ReLU(-\Delta TE_i)\right] \tag{14}$$

$$loss = L_{task} + \lambda_1 L_{path} + \lambda_2 (TE^{*} - TE)^2 + \lambda_3 H(g) \tag{15}$$

Where, λ_1, λ_2 and λ_3 are the weights for the regularization terms.L_{task} is the focal loss; L_{path} provides explicit enhancement rewards for key causal paths.The $(TE^{*} - TE)^2$ constrains the enhanced adjacency matrices to remain consistent with the initial TE structure, and $H(g)$ is the entropy regularization to improve graph robustness and stability.

4. Results

All experiments were conducted on a workstation equipped with an NVIDIA RTX 4090D GPU (24 GB) and an Intel Core i7-14700 CPU. The software environment consisted of Ubuntu 20.04, Python 3.8, PyTorch 2.0, and CUDA 11.8.All training was conducted on the GPU, and each complete run requiring approximately 5 h.

4.1. Data Description and procession

As one of the most common collision types, car-following collision were used to validate the effectiveness of the proposed framework (Wu et al., 2025). This study employed high-resolution vehicle trajectory data from the HighD dataset, collected by unmanned aerial vehicles (UAVs) at six different locations near Cologne, Germany, between 2017 and 2018. Vehicle trajectories were extracted from aerial imagery using advanced computer vision algorithms. The dataset includes 60 recordings, each comprising aerial video, metadata, and trajectory information (Wu et al., 2025). In this study, we selected a subset of 55 recordings, containing a total of 90,748 vehicle trajectories. Fig. 4 showed a six-lane bidirectional highway segment, where lanes 6, 7, and 8 correspond to direction 1 (east to west). The HighD dataset adopted a global image coordinate system, with the origin at the upper-left corner. Therefore, vehicles traveling in direction 1 exhibit negative velocity values due to the reversed coordinate orientation.

The trajectory data were first preprocessed to eliminate invalid records by removing points with negative velocity or missing values. Subsequently, car-following records were reconstructed by linking each vehicle to its corresponding preceding vehicle, with sequences divided into sub-sequences when temporal gaps exceeded two frames. Finally, risk events were identified using a Time-to-Collision (TTC) threshold: an event was labeled as dangerous if the minimum TTC fell below 3.0 s and the car-following behavior persisted for more than 100 frames, consistent with previous studies (Wu et al., 2025; Zhang et al., 2024). It should be noted that this case only serves to verify the theoretical effectiveness of the proposed framework based on predefined high-risk events.

In total, 240 dangerous car-following events were identified. As shown in Fig. 5, the minimum TTC values were mainly concentrated between 1.5 and 3.0 s, with a clear right-skewed distribution. A few events approached 1.0 s. Most events fell within a low dangerous range, with a higher frequency between 2.0 and 2.8 s. Event durations were also right-skewed, mostly between 5 and 12 s. A small number exceeded 25 s, with the longest lasting 38 s. These results indicate that dangerous car-following events are predominantly situated within a low risk range and exhibit widely varying durations. This provides a robust empirical foundation for subsequent causal modeling and consistent with the previous studies (Han et al., 2025).

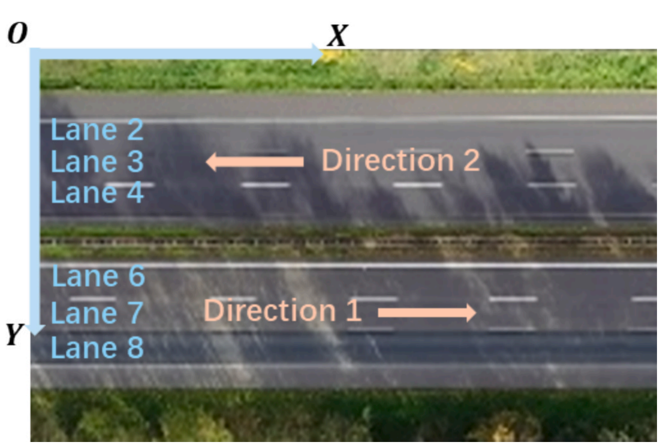


Fig. 4. Schematic of a Six-Lane Highway Segment.

4.2. Initial causal graph from TE analysis

In the causal discovery phase, a total of 240 hazardous car-following events were analyzed using the TE method to identify causal relations between traffic variables. For each event, eight traffic variables were considered following the previous settings (Zhang et al., 2024), including time-to-collision (*ttc*), inter-vehicle distance (distance), velocity difference (*v_diff*), acceleration difference (*acc_diff*), following vehicle speed (*v_fol*), preceding vehicle speed (*v_pre*), following vehicle acceleration (*acc_fol*) and preceding vehicle acceleration (*acc_pre*).

For every event, an 8 × 8 TE matrix was generated, resulting in 240 directed causal graphs for the following risk prediction component. To assess variable importance, TE values were aggregated to calculate feature importance scores. As shown in Fig. 6**(a)**, the most influential variables were *acc_pre, acc_fol*, and *v_pre*, followed by *acc_diff* and *v_diff*. These results indicate that both velocity-related and acceleration-related variables play critical roles in risk dynamics, with preceding vehicle behavior producing strong causal influence (Jin et al., 2025).

Beyond feature-level analysis, common causal edges across the 240 events were examined. As shown in Fig. 6**(b)**, the most frequent edges included *v_pre* → *distance* with 33 occurrences, *v_pre* → *ttc* with 28 occurrences and *v_fol* → *v_diff* with 24 occurrences. These relationships are consistent with traffic dynamics: the leading vehicle's speed directly affects inter-vehicle distance and TTC, whereas the following vehicle's speed determines velocity differences that drive traffic risk. These findings highlight key interactions between leading and following vehicles, providing informative priors for subsequent structural causal modeling and risk prediction.

4.3. Model performance

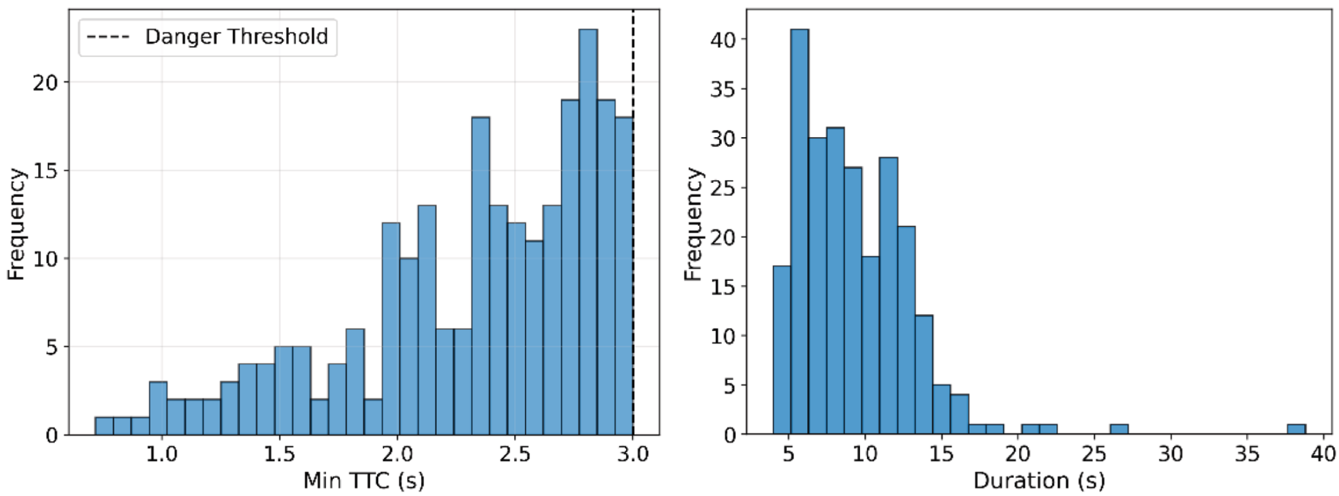
4.3.1. Model Parameter Configuration

A sliding window approach was employed to generate sequential training samples. Each of the 240 dangerous car-following events was segmented into overlapping sequences using a window size of 100 frames. The first 70 frames (with a stride of 10) served as input to predict the collision risk in the subsequent 30 frames. This process generated 5,884 training samples with an imbalanced distribution where hazardous samples constituted approximately 10% of the dataset. To ensure robust evaluation, five-fold stratified cross-validation was adopted, with each fold consisting of 70% training data, 15% validation data and 15% testing data.

Based on this data preparation, several training strategies were implemented to optimize model performance. A dynamic learning rate scheduling strategy was adopted to improve convergence stability. To alleviate the class imbalance, a focal loss function with label smoothing was applied to enable dynamic class weighting, allowing the model to focus more on hard-to-classify hazardous samples. Additionally, gradient accumulation was employed to stabilize optimization under limited batch sizes. All key hyperparameters were summarized in Table 1.

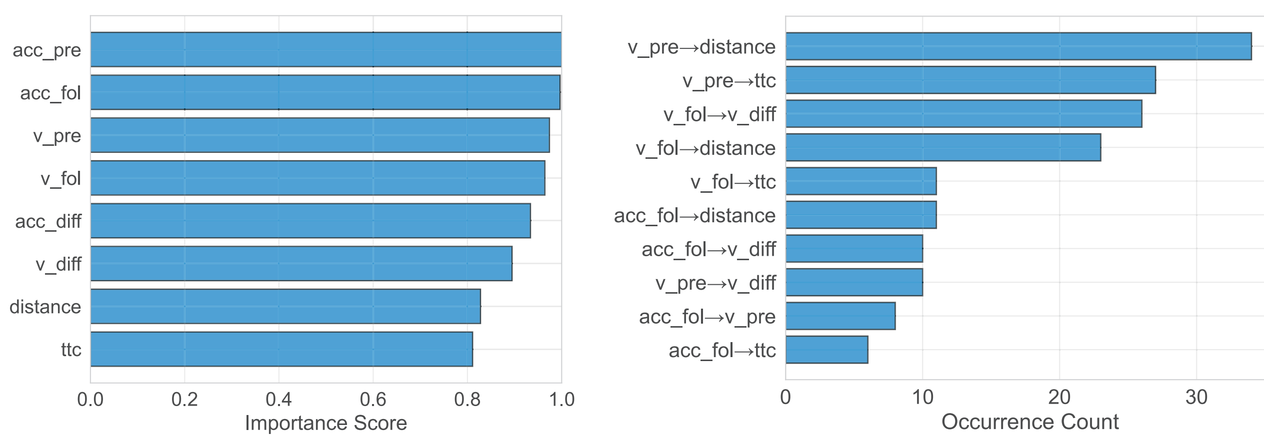
4.3.2. Model training and evaluation

Fig. 7 illustrated the training dynamics and performance of the GPT-2-LoRA model on the car-following risk prediction component. The solid curves show the average results across five cross-validation folds, and the shaded regions represent standard deviations, reflecting the model's stability under different data splits. Both training and validation losses decreased steadily and plateaued around epoch 150, indicating effective convergence without overfitting. The training loss declined from



(a) Min TTC distribution (b) Duration distribution

Fig. 5. Distribution analysis of dangerous events.



(a)Average feature importance scores (b) Most common causal edges

Fig. 6. Feature importance and causal relationship analysis based on 240 TE Results.

approximately 0.3 to 0.1, while the validation loss dropped to almost zero, demonstrating strong generalization capability.

In terms of classification performance, both the F1-score and accuracy increased rapidly during the first 30 epochs and then stabilized, reaching above 0.87 and 0.95 on the validation set. These results suggest that the model effectively balanced precision and recall, even under class imbalance. Furthermore, precision improved sharply in the early stages and remained stable between 0.83 and 0.85, indicating high confidence in hazardous event predictions and a low false-positive rate. Recall increased steadily and reached 0.91 after epoch 100, suggesting strong sensitivity to high-risk samples. Despite early fluctuations caused by class imbalance and noisy features, the model gradually adapted and enhanced its detection of minority-class instances. Moreover, the Area Under the Curve(AUC) exceeded 0.98 in the early epochs and remained consistently high throughout training, confirming robust discriminative performance across different decision thresholds.

Finally, the narrow shaded areas in later training stages indicate low variance across folds, reinforcing the model's stability. This is further supported by the close alignment between the training and validation curves, indicating strong generalization and robustness.

4.3.3. *Structural changes in the enhanced causal graph*

To reveal how causal enhancement restructures the graph of a high-risk event, event 128 was selected for comparative analysis. Fig. 8 showed the original TE-based causal graph and its enhanced version. In the original graph, the edge weights among variables were relatively uniform, with weak links among *acc_fol, v_fol* and risk variables. This showed a dispersed structure lacking a clear risk propagation path. After enhancement, weak links between non-essential variables were suppressed. For example, the edge weights of *acc_fol* → *acc_pre* and *acc_fol* → *v_pre* decreased by approximately 0.03, indicating a down-weighting of less informative connections. Meanwhile, the key causal paths related to traffic risk were strengthened. The edge weights of *acc_pre* → *ttc, v_pre* → *ttc, and acc_diff* → *ttc* increased by about 0.04, forming a clear forward-directed causal structure toward ttc. These changes highlight critical causal pathways and improve the interpretability of risk transmission in

Table 1
The key parameters of LLMG + ST model.

Models	Hyperparameters	Value
GPT2-LoRA	Embedding dimension	512
	Number of layers	8
	Number of attention heads	8
	LoRA rank	16
	LoRA scaling	32
	Max sequence length	600
	Initial learning rate	5×10^{-5}
	Dropout	0.15
LSTM	Hidden dimension	96
	Number of layers	2
	learning rate	9.6×10^{-5}
	Dropout	0.3
GAT	Number of graph layers	3
	Number of attention heads	6
	learning rate	6.4×10^{-5}
	Hidden dimension	32
	Output dimension	64
Training parameters	Weight decay	1×10^{-3}
	Batch size	32
	Gradient accumulation steps	2
	Early-stopping patience	20
	Max epochs	500
	Number of CV folds	5

dangerous car-following scenarios.

To furhter explore causal information while keeping the basic structure of the initial TE graph, we applied fine enhancements by GPT-2 to the initial graph. Fig. 9**(a)** summarized the differences in network density between the original and enhanced graphs across all 340 events. In the original graphs, density was concentrated between 0.80 and 0.87, with a median of 0.83. In contrast, the enhanced graphs showed a downward shift, with the median reduced to 0.77. Additionally, the variance increased, indicating greater topological diversity across samples.

Fig. 9**(b)** compared the edge weight distributions before and after enhancement. The results indicated that both distributions exhibited approximately normal shapes.Specifically, the enhanced distribution shifts slightly rightward, with the peak moving from 0.08 to 0.12. This suggested that the enhancement process improved the graph's representational capacity while largely preserving its original causal structure.

Table 2 showed the average changes in structural attributes during causal graph enhancement. The results indicated that most variables showed reductions in average in-degree and out-degree characters, indicating a pruning of redundant connections. For instance, the in-degree of *v_fol, acc_fol*, and *v_pre* decreased by 0.620, 0.280, and 0.849. In contrast, the in-strength of *ttc* increased by 0.232, enhancing its causal acceptability as a key dependent variable. The average out-strength of *distance* and *acc_pre* rose by 0.051 and 0.059, reflecting stronger outward influence on other variables within the graph.These changes revealed a clearer differentiation of node roles. The variable *ttc* served as a strengthened outcome node with greater incoming causal influence, while *distance, acc_pre, and v_diff* consolidated their positions as causal drivers with enhanced outgoing strength. As a result, the enhanced graph exhibited a more hierarchical causal structure.

Fig. 10 illustrated the enhanced edge weights(left)and the corresponding changes relative to the original graph(right). The enhanced graphs exhibited a bimodal distribution, with weights concentrated below 0.06 or above 0.15.This indicates a polarized distribution of connection strengths. Key causal paths were significantly strengthened, with *acc_pre* → *ttc* reaching 0.157, *v_pre* → *ttc* reaching 0.156, and *acc_diff* → *ttc* reaching 0.155, each increasing by about 0.06. These enhancements highlighted the dominant role of acceleration and velocity variables in determining the target risk factor. Meanwhile, some peripheral paths such as *acc_fol* → *acc_pre* and *acc_fol* → *acc_diff* were weakened, showing the tendency of the adjustment strategy to compress non-critical edges. Overall, the enhancement strategy effectively distinguished key causal pathways from weaker associations, resulting in a more focused and interpretable causal structure.

4.3.4. Model performance

Table 3 compared the performance of different models in real-time

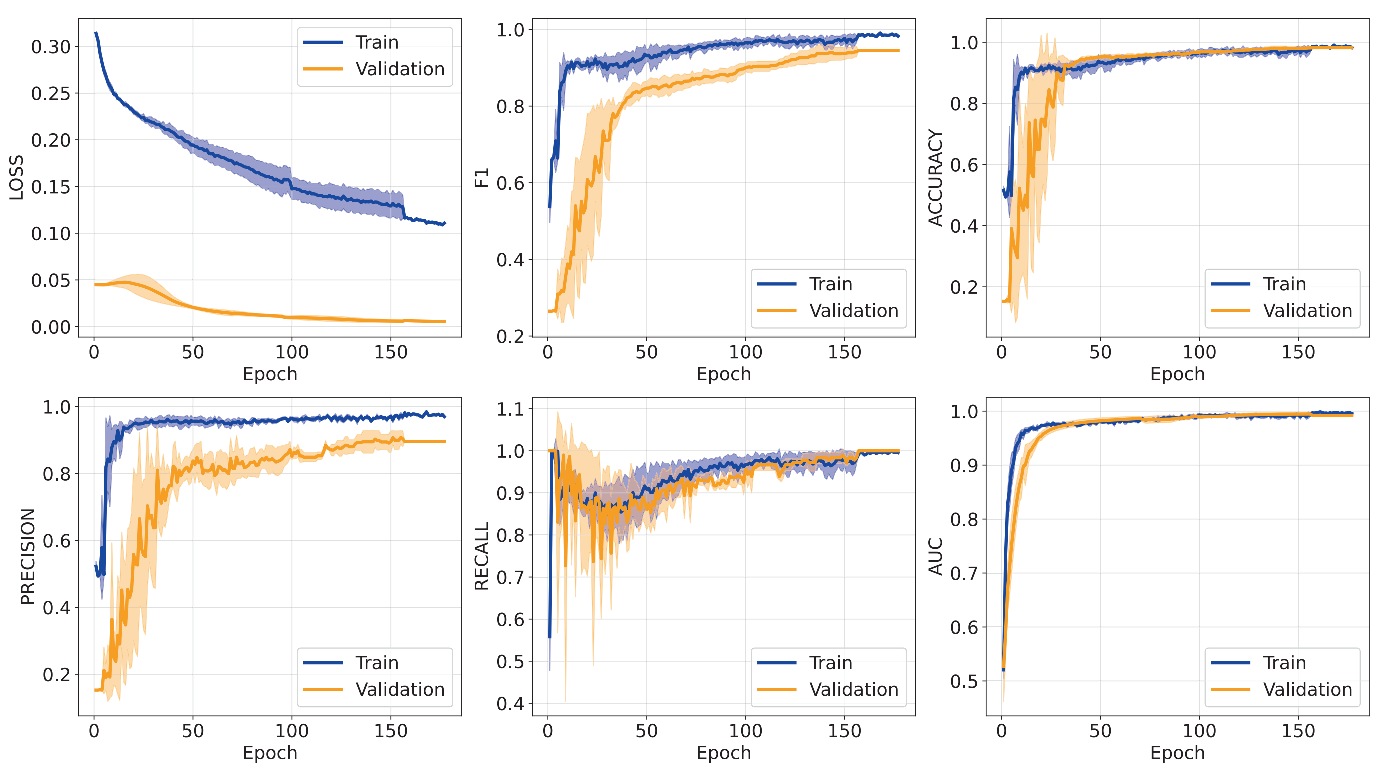


Fig. 7. Merged learning curves.

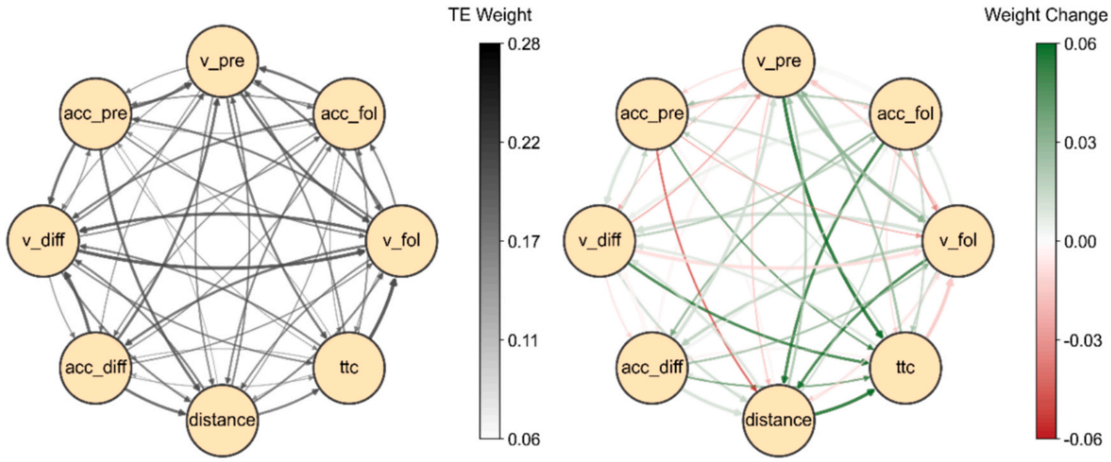


Fig. 8. Original causal graph and the enhanced graph for event 128.

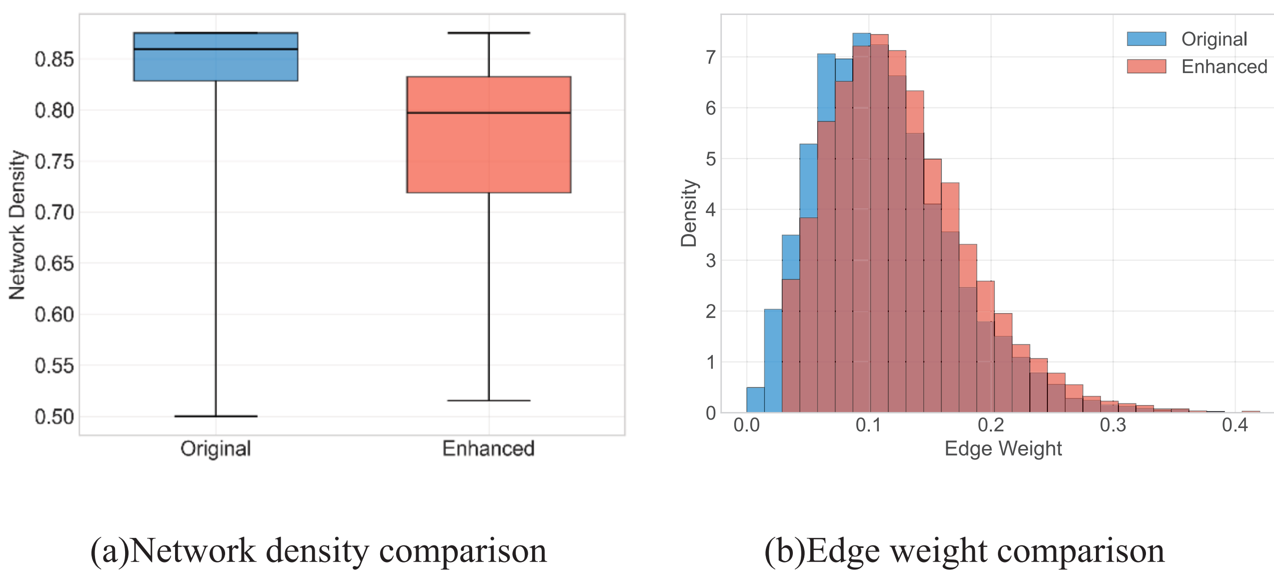


Fig. 9. Comparison between original and enhanced networks.

Table 2
Average changes in structural attributes.

Variable	In-Degree Change	Out-Degree Change	In-Strength Change	Out-Strength Change
v_fol	−0.620	−0.413	−0.009	0.046
acc_fol	−0.280	−0.383	−0.001	0.052
v_pre	−0.849	−0.228	−0.023	0.055
acc_pre	−0.112	−0.237	−0.004	0.059
v_diff	−0.306	−0.396	−0.003	0.056
acc_diff	−0.077	−0.525	−0.003	−0.016
distance	−0.599	−0.112	0.111	0.051
ttc	−0.004	−0.551	0.232	−0.008

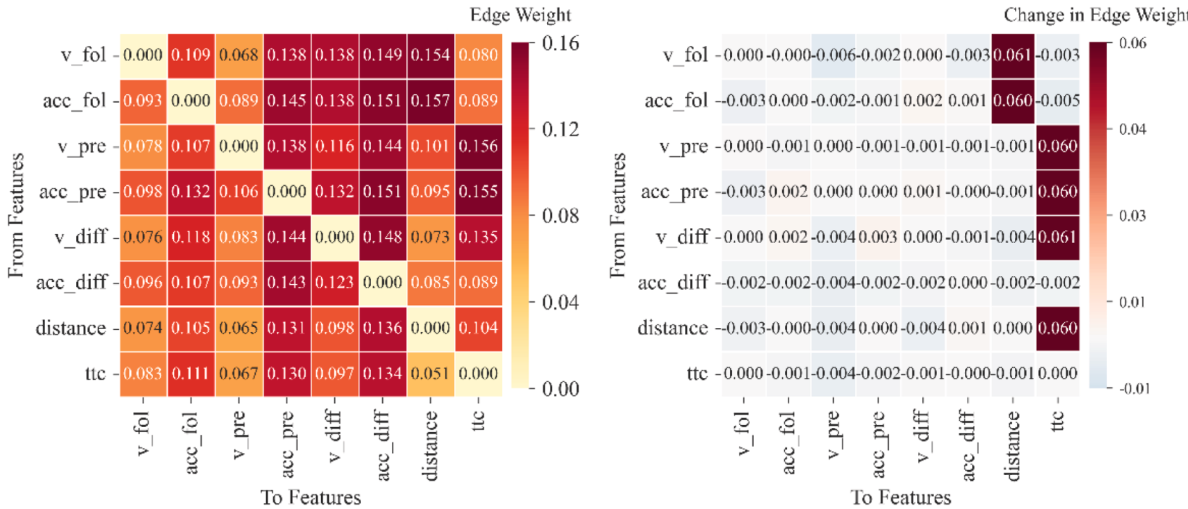
risk prediction. All baseline models were rigorously trained using identical data splits and imbalance-handling strategies as the proposed model. Traditional deep learning baselines, including LSTM, GRU and CNN-LSTM, achieved moderate accuracy between 0.800 and 0.814. However, their F1 scores remained low, ranging from 0.438 to 0.474, indicating limited performance in identifying hazardous events. Moreover, recall ranged from 0.495 to 0.545 while precision remains lower from 0.386 to 0.420, indicating a high false positive rate. The GCN model achieved higher recall at 0.594, suggesting better detection of risk events. However, this improvement came with lower accuracy at 0.791, reflecting reduced discrimination ability.

In contrast, the proposed LLMG + ST model significantly outperformed all baselines. It achieved an accuracy of 0.956, F1 score of 0.872, AUC of 0.985, precision of 0.832 and recall of 0.916. These results demonstrate its superior capacity to capture complex temporal patterns and causal dependencies, enabling more reliable collision risk prediction than traditional deep learning methods.

Beyond prediction accuracy, computational efficiency is critical for practical deployment. The inference latency of the proposed framework was evaluated to verify its real-time capability. Two deployment paradigms were assessed including offline pre-computed TE and fully online TE computation. The results indicate a total inference latency of 5–15 ms per sliding window, well within the 500 ms threshold required by Advanced Driver Assistance Systems (ADAS) and real-time traffic control. A detailed computational cost analysis under different deployment configurations is provided in Appendix B.

4.4. *Ablation study*

To further evaluate the effectiveness of LoRA fine-tuning and the entire causal graph enhancement in risk prediction, two ablation



(a) Enhanced Edge Weights (b) Edge Weight Changes

Fig. 10. Edge weight distributions before and after enhancement.

Table 3
Prediction performance of different models.

Model	Accuracy	F1 score	AUC	Precision	Recall
LLMG + ST	0.956	0.872	0.985	0.832	0.916
LSTM	0.814	0.474	0.814	0.420	0.545
GRU	0.812	0.448	0.817	0.410	0.495
CNN-LSTM	0.800	0.438	0.802	0.386	0.505
GCN	0.791	0.467	0.793	0.385	0.594

Table 4
Prediction performance of different experimental conditions.

Model	accuracy	F1 score	AUC	Precision	Recall
LLMG + ST	0.956	0.872	0.985	0.832	0.916
No LoRA	0.937	0.811	0.966	0.758	0.871
No Enhancement	0.934	0.804	0.972	0.751	0.866
No GAT	0.938	0.829	0.971	0.802	0.862
No LSTM	0.154	0.262	0.486	0.154	0.882

experiments were conducted based on the LLMG + ST model.

Experiment 1 (No-LoRA): The GPT-2 module was retained, but LoRA fine-tuning was removed. This model relied solely on the original, unfine-tuned GPT-2 to generate enhanced causal graphs. This experiment was designed to assess the specific contribution of LoRA fine-tuning to causal enhancement quality.

Experiment 2 (No-Enhancement): The entire GPT-2-based enhancement module was removed. The model used only the original TE causal graph and time series features, which were directly input into the ST component(LSTM-GAT) for prediction. This experiment evaluated the impact of causal graph enhancement on risk prediction performance.

Experiment 3 (No-GAT): The GAT module was removed.This model relied solely on the LSTM for sequence processing, utilizing the TE causal graph merely as basic statistical features. This experiment validated the contribution of GAT in capturing spatial causal relationships.

Experiment 4 (No-LSTM): The LSTM module was removed while retaining the GAT. The input temporal features.The model used only the GAT and the TE causal graph for prediction, with the input temporal features averaged over time.This experiment evaluated the impact of LSTM in capturing temporal dynamics for risk prediction.

As shown in Table 4, the full LLMG + ST model achieved the best performance, with an accuracy of 0.956, F1-score of 0.872, AUC of 0.985, precision of 0.832 and recall of 0.916. In contrast, the No-LoRA experiment showed lower performance, with F1 score decreasing to 0.811 and precision to 0.758, indicating that LoRA fine-tuning is essential for capturing causal relations. The No-Enhancement experiment, which completely removed the GPT-2-based enhancement component, relied on the static graph and temporal features within the loop-closed architecture. It achieved a recall of 0.866 and AUC of 0.972, both lower than the full model. Moreover, the No-GAT experiment obtained an F1 score of 0.829 and a recall of 0.862, suggesting that GAT plays an important role in capturing spatial causal dependencies within the enhanced graph. By contrast, the No-LSTM experiment showed the most severe degradation, with the accuracy falling to 0.154 and the AUC to 0.486. This pattern suggests that temporal features are critical for risk prediction. Overall, the LLMG + ST model significantly improved generalization and achieved a better balance between precision and recall in collision risk prediction.

5. Conclusions

This study proposed a closed-loop framework for real-time risk prediction that unifies causal discovery, LLM-based causal-graph enhancement and spatial–temporal prediction. By refining transfer-entropy graphs with GPT-2-LoRA and embedding them into the prediction model, this study demonstrated the potential of combining causal reasoning and semantic intelligence for reliable and explainable traffic risk assessment.

The findings showed that the enhanced TE graphs reinforced key causal pathways and reduced redundant links, thereby improving feature distinctions. Compared with baseline models, the proposed approach achieved higher accuracy, improved robustness across folds and greater interpretability. Ablation experiments further revealed that the causal graph enhancement with LoRA fine-tuning significantly improved the classification effectiveness and explanatory clarity, supporting practical deployment in safety-related interventions.

Based on these findings, several practical applications can be proposed to enhance traffic safety. By predicting collision risks through interpretable causal pathways, the targeted interventions, including adaptive warnings, dynamic headway control, and personalized driver alerts, can be implemented to reduce the possibility of rear-ending collisions. Moreover, the proposed framework may support the development of intelligent vehicle systems. Through the integration of LLM-based causal inference and spatio-temporal modeling, these systems can conduct precise countermeasures via analyzing the driving behavior between the targeted vehicles.

However, this study has several limitations. First, the causal graphs were constructed independently for each event and did not evolve with traffic dynamics, which limits their adaptability to changing conditions. Future work should therefore develop dynamic causal graph learning to capture temporal evolution. Second, the current framework relies mainly on basic driving variables from the ego and leading vehicle, while excluding external factors such as weather and road geometry, as well as broader multi-vehicle interactions. Incorporating multimodal data and extending the causal graph to model broader vehicle interactions would enhance the framework's generalizability. Third, although the proposed method has great potential for real-time risk prediction, it primarily focuses on short-term risk assessment within predefined dangerous scenarios. This limits its direct application to real-world driving, where dangerous periods must first be identified. Future work will incorporate safe driving datasets to learn safe traffic patterns, enabling direct risk prediction in practical deployment.

CRediT authorship contribution statement

Chenguang Li: Writing – original draft. **Helai Huang:** Writing – review & editing. **Hanchu Zhou:** Writing – review & editing, Supervision. **Fangying Li:** Writing – original draft. **Rui Zhou:** Writing – review & editing. **Liangjian Zhong:** Writing – original draft.

Declaration of competing interest

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Appendix A:. Description of prompt and semantic information

Causal Graph Enhancement Task.
Variables: {vars}.
Objective: Enhance causal graph for binary danger prediction
Variable Definitions.
v_fol: following vehicle speed − velocity of the car behind (m/s).
acc_fol: following vehicle acceleration − rate of speed change (m/s^2).
v_pre: leading vehicle speed − velocity of the front car (m/s).
acc_pre: leading vehicle acceleration − front car's speed change (m/s2).
v_diff: speed difference − relative velocity (v_fol minus v_pre).
acc_diff: acceleration difference − relative acceleration
distance: inter-vehicle distance − gap between cars (meters).
ttc: time to collision − seconds until crash if speeds maintained.
Current Graph Statistics.
Current density: {density:.3f}.
Strong causal edges (>0.1): { strong edges}.
Maximum causal strength: {max_te}.
Key Causal Paths.
Primary influences on ttc (traffic risk): { key edges}:
v_pre → ttc: {v_pre_to_ttc:.3f} (leading speed affects collision time).
acc_pre → ttc: {acc_pre_to_ttc:.3f} (leading braking impact).
distance → ttc: {dist_to_ttc:.3f} (gap to collision time).
v_diff → ttc: {v_diff_to_ttc:.3f} (speed difference effect).
acc_fol → distance: {acc_fol_to_dist:.3f} (following braking to gap).
v_fol → distance: {v_fol_to_dist:.3f} (following speed to gap).
Driving Knowledge.
If v_pre decreases → v_fol decreases (acc_fol < 0), distance tends to decrease initially.
distance ↓ → ttc ↓; v_diff ↑ (follower faster) → risk ↑.
acc_pre < 0 strengthens edges driving ttc decrease.
Highway Context.
Dataset: highD (German highway), same-direction traffic; weather/grade unknown.
Safety Rules.
TTC < 3 s indicates high risk.
Safe following: time_headway ≥ 2 s.
Comfortable deceleration: $|a| \leq 3$ m/s^2.
Enhancement Strategy.
Identify and strengthen critical causal paths for danger prediction.
Focus on causal chains: acceleration → speed → distance → ttc.
Output: Refined causal weights for improved precision and F1 score.

Appendix B. Computational cost and analysis

(1) Offline Pre-computed TE.

Vehicle Pairs	TE Cache Loading	Model Inference	Total Latency	Real-time (<500 ms)
1	0.13 ms	4.71 ms	4.84 ms	✓
4	0.52 ms	7.12 ms	7.64 ms	✓
8	1.04 ms	8.52 ms	9.56 ms	✓
16	2.08 ms	12.74 ms	14.82 ms	✓

(2) Online TE using Full Computation

Vehicle Pairs	TE Computation	Model Inference	Total Latency	Real-time (<500 ms)
1	6.9 ms	4.71 ms	11.6 ms	✓
4	31.4 ms	7.12 ms	38.5 ms	✓
8	67.2 ms	8.52 ms	75.7 ms	✓
16	136.6 ms	12.74 ms	149.3 ms	✓

Data availability

Data will be made available on request.

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